

Biochemistry I, Chemistry NNNN
Fall 2027

Instructor: Md. Sharear Saon Room: Somewhere Email: sharearsaon@outlook.com Phone: (314)500-3617	Class schedule: MWF 10:00 am – 10:50 am Class location: Somewhere Office hours: Friday 3:00 pm – 4:00 pm Office hours location: Somewhere or zoom
---	---

Course Description

This course introduces the chemical principles that govern the structure, function, and interactions of biological molecules. Topics include water and biological thermodynamics; amino acids and proteins; carbohydrates, lipids, and membranes; nucleotides and nucleic acids; recombinant DNA methods; and enzyme kinetics, mechanisms, and regulation. Students will connect molecular structure to biological function, solve quantitative biochemical problems, and interpret experimental data. Prior coursework in general chemistry and organic chemistry is recommended.

Goals

1. Connect chemical principles to the structures and behavior of biological molecules.
2. Relate biomolecular structure, dynamics, and interactions to biological function.
3. Develop quantitative reasoning through thermodynamics, binding, and enzyme kinetics.
4. Use biochemical evidence to explain mechanisms and evaluate scientific conclusions.

Objectives

1. Students will explain how water, noncovalent interactions, and thermodynamics shape biological systems.
2. Students will compare amino-acid properties and relate protein sequence to three-dimensional structure and function.
3. Students will analyze enzyme kinetics, inhibition, catalytic mechanisms, and regulation using quantitative models.
4. Students will describe the structures and cellular roles of carbohydrates, lipids, membranes, nucleotides, and nucleic acids.
5. Students will interpret biochemical graphs, molecular structures, and experimental datasets.
6. Students will solve biochemical problems and communicate evidence-based conclusions clearly.

Required Course Materials

1. *Biochemistry, 7th Edition*
Authors: Reginald H. Garrett and Charles M. Grisham
ISBN-13: 978-0357728451
ISBN-10: 0357728459
2. Non-programmable scientific calculator
3. iClicker 2
You can purchase the textbook and iClicker through the bookstore. Register your iClicker in Canvas by September 7th.
4. Canvas account through MySLU tools.
5. Computer with internet connection to access course materials.

Technology used in the course

- PowerPoint
 - Lecture slides and guided notes will be presented in class and posted in Canvas.
- iClicker
 - Using conceptual and quantitative questions to support active learning.
- Canvas
 - Posting course materials (announcements, grades, assignments, feedback)
 - Posting questions (anyone can post and respond to another post)
 - Interactive reading assignments (highlight, ask questions, and respond to peers).
- Google Drive
 - Protein structures, biochemical datasets, and additional readings will be shared through Google Drive links.
 - Google Sheets can be used for collaborative data-analysis activities in class.

Assessments

- 1. iClicker questions (50 points, 5% of the total)**
 - Each class meeting will include at least one conceptual or quantitative iClicker question.
 - Questions may be used to revisit prior knowledge, predict an outcome, or interpret biochemical evidence.
 - Students will discuss selected questions with peers before responding again.
 - Students who miss class will not receive iClicker points for that day, except when an approved accommodation applies.
- 2. Online quizzes (100 points, 10% of the total)**
 - Ten short quizzes will be assigned across the major course modules.
 - Each quiz will include conceptual, structural, and quantitative questions.
 - Quizzes will be posted in Canvas before the related class meeting or module deadline.
 - Each quiz is worth 10 points.
 - Due dates will be announced in class and through Canvas notifications.
- 3. In-class activities (100 points, 10% of the total)**
 - There will be ten individual or small-group activities worth 10 points each.
 - Activities may include protein-structure interpretation, pathway mapping, or enzyme-kinetics problems.
 - Students will receive credit for a complete, evidence-based attempt and active participation.
 - Selected activities will conclude with a whole-class discussion and feedback.
- 4. Reading and reflection assignments (50 points, 5% of the total)**
 - Five reading and reflection assignments will be posted in Canvas (10 points each).
 - Assignments may draw on the textbook, a short research article, or a biochemical case study.
 - Students will identify a central claim, supporting evidence, and one remaining question.
- 5. Data-analysis quizzes (100 points, 10% of the total)**
 - There will be five short data-analysis quizzes worth 20 points each.
 - Each quiz will ask students to interpret graphs, molecular structures, or experimental results.

- Quizzes will follow major units and will emphasize reasoning rather than memorization.
 - Question formats may include calculations, multiple choice, short answers, and annotated figures.
- 6. Homework assignments (150 points, 15% of the total)**
- There will be five homework assignments worth 30 points each.
 - Assignments will combine quantitative problems, molecular interpretation, and short explanations.
 - Each assignment will be posted in Canvas at least one week before its due date.
 - Students must submit their work in Canvas by the stated deadline.
- 7. Exams (450 points, 45% of the total)**
- There will be four unit exams and one cumulative final exam.
 - Each unit exam is worth 80 points; the cumulative final exam is worth 130 points.
 - Exams may include multiple-choice questions, calculations, molecular structures, data interpretation, and written explanations.
 - Written work may receive partial credit when the reasoning is clear and scientifically supported.
 - Unit exams will fit within one class period; the final exam will follow the university exam schedule.
 - Phones, smart watches, and other unauthorized devices must be stored during exams.
 - Students needing testing accommodations should coordinate with Disability Services and the instructor before the exam.
 - The final exam will be cumulative, with emphasis on connections among biomolecular structure, energetics, and function.
 - If a student's final-exam percentage exceeds the lowest unit-exam percentage, the lowest unit-exam score will be replaced by the average of those two percentages.

Grades

In the final result, letter grade will be assigned using the following schema.

Letter grade	Point range	%	Letter grade	Point range	%
A	900-1000	90.00-100.00%	C+	650-699	65.00-69.99%
A-	850-899	85.00-89.99%	C	600-649	60.00- 64.99%
B+	800-849	80.00-84.99%	C-	550-599	55.00- 59.99%
B	750-799	75.00- 79.99%	D	450-549	45.00 - 54.99%
B-	700-749	70.00- 74.99%	F	<450	<45.00%

Academic Integrity

Academic integrity is honest, truthful and responsible conduct in all academic endeavors. The mission of Saint Louis University is "the pursuit of truth for the greater glory of God and for the service of humanity." Accordingly, all acts of falsehood demean and compromise the corporate endeavors of teaching, research, health care, and community service through which SLU embodies its mission. The University strives to prepare students for lives of personal and professional integrity and therefore regards all breaches of academic integrity as matters of serious concern.

The governing University-level Academic Integrity Policy was adopted in Spring 2015, and can be accessed on the Provost's Office website (<http://www.slu.edu/provost/policies.php>).

Additionally, each SLU college, school and center has adopted its own academic integrity policies, available on their respective websites. All SLU students are expected to know and abide by these policies, which detail definitions of violations, processes for reporting violations, sanctions, and appeals. Please direct questions about any facet of academic integrity to your faculty, the chair of the department of your academic program, or the dean/director of the college, school or center in which your program is housed.

Specific College of Arts and Sciences Academic Honesty Policies and Procedures may be found here (<http://www.slu.edu/arts-and-sciences/student-resources/academic-honesty.php>).

Title IX

Saint Louis University and its faculty are committed to supporting our students and seeking an environment that is free of bias, discrimination, and harassment. If you have encountered any form of sexual misconduct (e.g. sexual assault, sexual harassment, stalking, domestic or dating violence), we encourage you to report this to the University. If you speak with a faculty member about an incident of misconduct, that faculty member must notify SLU's Title IX coordinator, Anna R. Kratky (DuBourg Hall, room 36; anna.kratky@slu.edu; 314-977-3886) and share the basic facts of your experience with her. The Title IX coordinator will then be available to assist you in understanding all of your options and in connecting you with all possible resources on and off campus.

If you wish to speak with a confidential source, you may contact the counselors at the University Counseling Center at 314-977-TALK. To view SLU's sexual misconduct policy and for resources, please visit the following web addresses:
<http://www.slu.edu/about/safety/pdfs/sexual-misconduct-policy-version7.0.pdf>.

Student Success Center

In recognition that people learn in a variety of ways and that learning is influenced by multiple factors (e.g., prior experience, study skills, learning disability), resources to support student success are available on campus. The Student Success Center assists students with academic and career related services, is located in the Busch Student Center (Suite, 331) and the School of Nursing (Suite, 114). Students who think they might benefit from these resources can find out more about:

- Course-level support (e.g., faculty member, departmental resources, etc.) by asking your course instructor.
- University-level support (e.g., tutoring services, university writing services, disability services, academic coaching, career services, and/or facets of curriculum planning) by visiting the Student Success Center.

Disability Services Academic Accommodations

Students with a documented disability who wish to request academic accommodations are encouraged to contact Disability Services to discuss accommodation requests and eligibility requirements.

Please contact Disability Services, located within the Student Success Center, at disability_services@slu.edu or 314.977.3484 to schedule an appointment. Confidentiality will be observed in all inquiries.

Once approved, information about the student's eligibility for academic accommodations will be shared with course instructors via email from Disability Services and viewed within Banner via the instructor's course roster.

Course Schedule

Week	Date	Day	Lecture content
Week 1	August 30, 2027	Monday	Course introduction; Chapter 1: The Chemical Logic of Life
Week 1	September 1, 2027	Wednesday**	Chapter 1 (Cont): Molecular Interactions and Biochemical Evidence
Week 1	September 3, 2027	Friday**	Chapter 2: Water and Noncovalent Interactions
Week 2	September 6, 2027	Monday	Labor Day (No class)
Week 2	September 8, 2027	Wednesday	Chapter 2 (Cont): Acids, Bases, pH, and Buffers
Week 2	September 10, 2027	Friday	Chapter 3: Biological Thermodynamics
Week 3	September 13, 2027	Monday**	Chapter 3 (Cont): Free Energy, Equilibria, and Coupled Reactions
Week 3	September 15, 2027	Wednesday	Chapter 4: Amino Acids and Peptide Bonds
Week 3	September 17, 2027	Friday*	Chapter 4 (Cont): Ionization, Titration Curves, and Peptides
Week 4	September 20, 2027	Monday**	Chapter 5: Protein Primary Structure and Function
Week 4	September 22, 2027	Wednesday	Chapter 5 (Cont): Protein Sequencing and Proteomics
Week 4	September 24, 2027	Friday	Chapter 5 (Cont): Sequence Variation and Function
Week 5	September 27, 2027	Monday	<u>Exam 1</u>
Week 5	September 29, 2027	Wednesday	Exam 1 review; Chapter 6: Higher-Order Protein Structure
Week 5	October 1, 2027	Friday**	Chapter 6 (Cont): Secondary, Tertiary, and Quaternary Structure
Week 6	October 4, 2027	Monday	Chapter 6 (Cont): Protein Folding, Stability, and Dynamics
Week 6	October 6, 2027	Wednesday*	Chapter 6 (Cont): Methods for Studying Protein Structure
Week 6	October 8, 2027	Friday**	Chapter 7: Carbohydrates and Glycoconjugates
Week 7	October 11, 2027	Monday	Chapter 7 (Cont): Recognition and Cell-Surface Chemistry
Week 7	October 13, 2027	Wednesday**	Chapter 8: Lipids
Week 7	October 15, 2027	Friday	Chapter 8 (Cont): Lipid Structure, Assembly, and Signaling

Week	Date	Day	Lecture content
Week 8	October 18, 2027	Monday	Chapter 9: Membranes and Membrane Transport
Week 8	October 20, 2027	Wednesday	Chapter 9 (Cont): Channels, Carriers, and Active Transport
Week 8	October 22, 2027	Friday	Fall break (No class)
Week 9	October 25, 2027	Monday	<u>Exam 2</u>
Week 9	October 27, 2027	Wednesday	Exam 2 review; Chapter 10: Nucleotides and Nucleic Acids
Week 9	October 29, 2027	Friday* **	Chapter 10 (Cont): Nucleotide Chemistry and Biological Roles
Week 10	November 1, 2027	Monday	Chapter 11: Structure of Nucleic Acids
Week 10	November 3, 2027	Wednesday	Chapter 11 (Cont): DNA and RNA Structure, Dynamics, and Recognition
Week 10	November 5, 2027	Friday**	Chapter 12: Recombinant DNA, Gene Editing, and Synthetic Biology
Week 11	November 8, 2027	Monday	Chapter 12 (Cont): Experimental Design and Molecular-Biology Tools
Week 11	November 10, 2027	Wednesday	Data-analysis activity: Evaluating a Gene-Editing Experiment
Week 11	November 12, 2027	Friday	Chapter 13: Enzymes - Kinetics and Specificity
Week 12	November 15, 2027	Monday	Chapter 13 (Cont): Michaelis-Menten Kinetics and Inhibition
Week 12	November 17, 2027	Wednesday*	Enzyme-kinetics problem solving and data interpretation
Week 12	November 19, 2027	Friday**	Chapter 14: Enzyme Mechanisms and Catalytic Strategies
Week 13	November 22, 2027	Monday	<u>Exam 3</u>
Week 13	November 24, 2027	Wednesday	Thanksgiving break (No class)
Week 13	November 26, 2027	Friday	Thanksgiving break (No class)
Week 14	November 29, 2027	Monday	Chapter 15: Enzyme Regulation
Week 14	December 1, 2027	Wednesday	Chapter 15 (Cont): Allosteric and Covalent Regulation
Week 14	December 3, 2027	Friday**	Chapter 16: Molecular Motors
Week 15	December 6, 2027	Monday	Chapter 16 (Cont): Chemomechanical Coupling and Cellular Motion
Week 15	December 8, 2027	Wednesday*	Integrative case study: From Molecular Structure to Cellular Function
Week 15	December 10, 2027	Friday	Protein structure-function project presentations and course synthesis
Week 16	December 13, 2027	Monday	<u>Exam 4</u>
Week 16	December 15, 2027	Wednesday	Cumulative review and concept integration
Week 16	December 17, 2027	Friday	No class
	December 20, 2027	Monday	<u>Final Exam</u>

* Assignment will be posted in Canvas

** Quiz (covered the chapter that was just finished) day